Department of Economics

MSc Extended Essay Cover Sheet

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TITLE	**Financial Repression, Debt Structure and Capital Accumulation in a New Keynesian Framework (Structured Research Assignment)**

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Double spaced, typed. Maximum 6,000 words (as a guideline this is 14-21 double spaced pages). Abstract, footnotes, references and appendices do not count toward the word count, provided such additions are brief and do not contain information that rightly belongs in the body of the essay. Equations are included in the word count and counted as the page equivalent (i.e., as the number of words that would occupy the same amount of space in text). Examiners are recommended to penalise excessively long essays (+ 10% of word count) by adjusting the marks by 6000/x, where x is the word count. Pages are to be numbered.*

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Financial Repression, Debt Structure and Capital Accumulation in a New Keynesian Framework

Structured Research Assignment

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Abstract

This essay analyses financial repression in a basic New Keynesian framework with government debt. Fiscal shocks force the planner to reduce the real value of the debt stock by imposing real losses on current and future bondholders, with the optimal mix of the two channels depending on the parameters of the economy. A debt structure consisting only of either one-period bonds or perpetuities is found to reduce the welfare loss to a minimum under the baseline calibration. Finally, the effects of financial repression on investment are considered by enriching the model with endogenous capital accumulation as in an RBC framework. Capital accumulation is found to limit the ability of the planner to manipulate ex-ante real interest rates, making longer-term debt structures more attractive vis-à-vis short-term debt.

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1 Introduction

In the aftermath of the COVID-19 pandemic, public debt levels are at all-time highs in most developed economies (OECD, 2022), raising concerns about the sustainability of the burden on government finances. Debt-to-GDP ratios were generally already high before the pandemic, but the environment of low interest rates and high demand for safe assets helped allay fears about unsustainable debt levels, since investors were apparently ready to absorb any amount of government bonds thrown at them. Now, however, rising inflation

raises the question: is the pre-pandemic environment gone for good? Will future fiscal shocks need to be paid for the old-fashioned way, via higher net primary surpluses? Will bond markets refuse to foot the bill when the next recession comes around, forcing painful adjustments in public finances at the worst point in the business cycle? Are there any alternatives available to policy makers? If there are, what are their costs?

In the past, governments have frequently resorted to financial repression to reduce the debt stock after wars and other large shocks to the public budget (Reinhart et al., 2015). In what follows, financial repression is understood to mean a policy that deliberately imposes negative real returns on holders of public debt instruments, thus reducing the burden for the government. This essay tries to understand *the cost* this imposes on the economy by introducing public debt and fiscal shocks into a simple representative-agent New Keynesian model. First, the optimal (Ramsey) financial repression policy is derived analytically, and then simulations of the calibrated model are run to understand its quantitative effects on aggregate welfare. The main result of this exercise relates to the term structure of government debt, suggesting that welfare losses are minimised when governments exclusively issue either very short-term debt or perpetuities. Lastly, the standard New Keynesian model is enriched with endogenous capital accumulation to study the interaction between financial repression and investment.

The rest of this essay is structured as follows. Section 2 focuses on the basic New Keynesian model with fiscal shocks, with subsection 2.2 presenting the analytical solution to the optimal policy (Ramsey) problem, 2.3 analysing

the results of simulating the calibrated model and 2.4 studying the optimal choice of debt structure. Section 3 introduces capital into the New Keynesian model, with subsection 3.3 reporting the results of numerical simulation of the enriched model and 3.4 commenting on the qualitative effects of financial repression on welfare. Finally, section 4 concludes.

2 Financial repression in the basic New Keynesian model

2.1 Log-linearised model

Our log-linearised New Keynesian model consists of two blocks of equations. First, we have the standard New Keynesian Phillips Curve (equation 1), and the Euler equation or dynamic IS curve (equation 2):

$$\pi_t = \beta \mathbb{E}_t(\pi_{t+1}) + \kappa y_t \tag{1}$$

$$y_t = \mathbb{E}_t(y_{t+1}) - \sigma r_t \tag{2}$$

See Galí (2015) for a derivation of these relationships from scratch. The second block is the public debt side of the model:

$$r_t = \mathbb{E}_t(R_{t+1}) \tag{3}$$

$$j_t = \beta\delta \mathbb{E}_t(j_{t+1}) + (1 - \beta\delta) i_t \tag{4}$$

$$R_t = j_{t-1} - \pi_t - \frac{\beta\delta}{1-\beta\delta}(j_t - j_{t-1}) \tag{5}$$

$$\beta d_t = d_{t-1} + R_t + \beta\phi_t \tag{6}$$

Equation 3 defines the ex-ante real interest rate as the expected ex-post real return on government bonds (the only asset in our economy). Equation 4 gives the relationship between the bond yield, j_t, and the one-period nominal interest rate, i_t (note that this purely a definition, since no asset pays the one-period rate), and can be thought of as the expectations hypothesis of the term structure of interest rates. Equation 5 defines the ex-post real return. Finally, equation 6 is the government's budget identity.

For simplicity, the only exogenous shocks in our model are the fiscal shocks, ϕ_t. In the absence of these shocks, it would be optimal to set both inflation and the output gap equal to zero in every period (the 'divine coincidence'). When the shocks are non-zero, the setup pretty much forces the government to deviate from the zero-inflation rule, because otherwise the real debt stock spirals out of control. Intuitively, in this economy there is zero output growth and a positive real interest rate in steady state ($r > g$), so any amount of real debt above the sustainable level will increase exponentially as interest accumulates. More formally, we can see this from the government's budget identity. Assuming inflation and the real interest rate remain at their steady state levels, and that there is a positive fiscal shock at time t_0:

$$d_{t_0} = \phi_{t_0}$$

$$d_{t_0+1} = \frac{1}{\beta} d_{t_0} = \frac{1}{\beta} \phi_{t_0}$$

$$d_{t_0+2} = \frac{1}{\beta} d_{t_0+1} = \left(\frac{1}{\beta}\right)^2 \phi_{t_0}$$

$$\dots$$

$$d_{t_0+h} = \left(\frac{1}{\beta}\right)^h \phi_{t_0}$$

Since $\beta < 1$, the real debt stock explodes on its own after a positive fiscal shock unless the government changes policy. An alternative way to see this would be to iterate the budget constraint forward, impose the transversality condition and set $d_{t_0-1} = 0$ (i.e., we begin from steady state), which gives us the following intertemporal budget constraint:

$$R_{t_0} + \beta \sum_{h=0}^{\infty} \beta^h \mathbb{E}_{t_0}(r_{t_0+h}) = -\beta \phi_{t_0} \tag{7}$$

Thus, a positive fiscal shock happening today must be offset either by a negative ex-post return today or by negative ex-ante real interest rates in the future. The trade-off the government faces in this model is that attempting to reduce the ex-ante real interest rate will induce higher spending and inflation, which generates welfare losses due to relative price distortions, whereas using the ex-post return channel by generating current inflation will induce positive output gaps.

2.2 Ramsey problem

The Ramsey planner minimises the following welfare-based[1] loss function subject to equations 1 through 6:

$$L_t = \mathbb{E}_t \sum_{h=0}^{\infty} \beta^h \frac{1}{2} (\frac{\varepsilon}{\nu\xi} \pi_{t+h}^2 + \gamma y_{t+h}^2) \tag{8}$$

The optimal policy can be characterised analytically by the following result:

Proposition 1 *Let $\{\pi_{t+h}, y_{t+h}\}_{h=0}^{\infty}$ minimise (8) subject to (1) through (6). Then, for all $h \geq 0$:*

$$\pi_{t+h} = \frac{\nu\xi}{\varepsilon} (\lambda_{t+h}^A - \lambda_{t+h-1}^A + \lambda_{t+h}^E) \tag{9}$$

$$y_{t+h} = \gamma^{-1} (\frac{1-\beta\delta}{\sigma} \lambda_{t+h}^E - \frac{1-\beta}{\sigma} \lambda_{t+h}^F - \kappa \lambda_{t+h}^A) \tag{10}$$

where $\left\{\lambda_{t+h}^A, \lambda_{t+h}^E, \lambda_{t+h}^F\right\}_{h=0}^{\infty}$ are the optimal values of the Lagrange multipliers associated with constraints (1), (5) and (6), respectively, and $\lambda_{t-1}^A = \lambda_{t-1}^E = \lambda_{t-1}^F = 0$. Also, for $h \geq 0$, the multipliers satisfy:

$$\lambda_{t+h}^E = \delta \lambda_{t+h-1}^E + \chi \phi_{t+h} \tag{11}$$

$$\lambda_{t+h}^F = \lambda_{t+h-1}^F + \chi \phi_{t+h} \tag{12}$$

$$\lambda_{t+h}^A = a_1 \lambda_{t+h-1}^A + a_2 \lambda_{t+h}^E + a_3 \lambda_{t+h}^F \tag{13}$$

[1] A derivation of a loss function of this form can be found in Chapter 4 of Galí (2015).

where a_1, a_2, a_3, and χ are known functions of the parameters of the model, given in Appendix A.

For a proof by construction of this result, see Appendix A.

Proposition 1 pins down the optimal path of inflation and the output gap, which are complicated functions of current and lagged values of the fiscal shocks. However, for certain values of the duration parameter δ, the law of motion of inflation takes a simpler form.

Consider first the case $\delta = 0$, which corresponds to the case of one-period debt. In this case, it can be shown that inflation behaves as a stationary ARMA(1, 1) process:

$$\pi_t = a_1 \pi_{t-1} + \frac{\nu \xi \chi}{\varepsilon} \left[(1 + a_2 + a_3)\phi_t - (a_1 + a_2)\phi_{t-1} \right] \tag{14}$$

For $\delta = 1$ (which can be interpreted as the perpetuity case), inflation becomes a random walk:

$$\pi_t = \pi_{t-1} + \frac{\nu \xi \chi}{\varepsilon} \phi_t \tag{15}$$

2.3 Calibration and simulation

Our economy has seven fundamental parameters:

- The discount rate, β. Since the steady state real interest rate in this model is $1 + \bar{r} = \beta^{-1}$, I calibrate the discount factor so that it is

consistent with a 2% annual real interest rate. With a time period equal to a month, this is approximately equivalent to $\beta = \left(\frac{1}{1.02}\right)^{\frac{1}{12}} \approx 0.9984$.

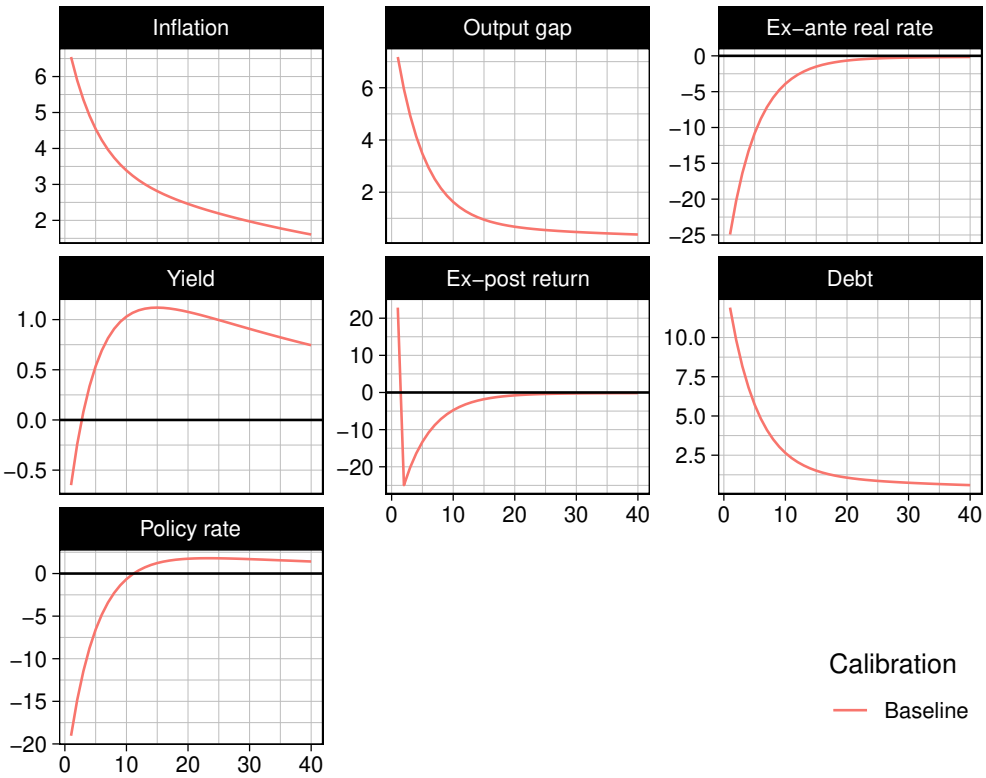
- The elasticity of intertemporal substitution (EIS), σ. There is substantial uncertainty in the literature about what the correct calibration for σ is: estimates of the elasticity range from close to 0 to more than 10 (Havranek et al., 2015). In principle I use a value of $\sigma = 0.594$, which according to Havranek et al. is the mean estimated EIS for the US economy, but values close to 0 and 2 are also employed as a robustness check.

- The price elasticity of demand, ε. In our model, this parameter is linked to the gross markup over marginal cost, which in steady state is $\frac{\varepsilon}{\varepsilon-1}$. Picking $\varepsilon = 6$ implies a steady-state average markup of 1.2, which is consistent with recent microeconomic evidence (see Barkai, 2020 and Basu, 2019).

- The Frisch elasticity of labour supply, ψ. Here I use a value of $\psi = 2.84$, which is the average of several macroeconomic estimates of the wage elasticity of labour supply (Chetty et al., 2011), although micro evidence tends to find much lower elasticities. Using the micro elasticity is likely to lead to unrealistically low volatility of aggregate employment in our model, but as a robustness check, I also run simulations using Chetty et al.'s recommended value, $\psi = 0.75$.

- The elasticity of output with respect to labour input, $1 - \alpha$. In steady state and assuming there is a wage-bill subsidy that offsets the ineffi-

Figure 1: Baseline calibration



ciency caused by monopolistic competition, this should be equal to the labour share of income, which is typically assumed to be around $\frac{2}{3}$ in advanced economies. Thus, a value $\alpha = \frac{1}{3}$ is used throughout.

- The fraction of firms that cannot adjust prices in a given period, θ. Nakamura and Steinsson (2008) report median price durations of 8 to 11 months, and the expected time between price changes in the Calvo staggered-price-adjustment framework is $\frac{1}{1-\theta}$, which suggests values of the parameter between $\theta = \frac{7}{8}$ and $\theta = \frac{10}{11}$. In my baseline calibration, $\theta = 0.9$, but I also experiment with very sticky and very flexible prices

to get a better intuition about the channels underlying optimal policy.

- The debt duration parameter, δ. In steady state, keeping the quantity of real debt d_t at a constant level requires the government to refinance a fraction $1-\delta$ of the debt stock each period. Thus, one can use the percentage of debt maturing in the next 12 months as a rough estimate of the long-term rollover ratio and then back out an implied value of δ. For the year 2022, G7 economies are expected to roll over 24% of their debt stock (OECD, 2022). Extrapolating to monthly periods, this implies a value of $1-\delta = \frac{0.24}{12} = 0.02$, or $\delta = 0.98$. As a check, note that δ is also related to the steady state value of the Macaulay duration of government debt in our model. The implied duration for $\delta = 0.98$ and $\beta = 0.9984$ is $\frac{1}{1-\beta\delta} \approx 46.26$ or 3.86 years, which is a reasonable value for developed economies.

For the baseline calibration, the impulse responses of inflation and the remaining endogenous variables to a fiscal shock equal to 10% of the steady state debt stock are given in figure 1. The Ramsey policy involves substantial inflation for a prolonged period: on impact, inflation jumps to an annualized rate of 6.54% and declines slowly, staying as high as 1.37% after 5 years. Somewhat counter-intuitively, the debt stock jumps by *more* than 10%, because the initial decline in nominal interest rates provides a windfall gain to bondholders: the nominal price of long-term debt is inversely related to the bond yield, and when prices are sticky, the initial rise in inflation is not large enough to offset this effect. Subsequently, persistence in inflation starts biting into the real return on bonds from the second period onwards, driving

down the debt stock towards the steady state level.

An evident weakness of the model is that the initial reduction in the policy rate (which the central bank implements to achieve the desired inflation) is extremely large (-19.04%). Even though the impulse responses should be interpreted as deviations from steady states values, the central bank is all but certain to run against the Effective Lower Bound (ELB) when trying to implement such an extreme policy. Studying how the ELB affects the choices of the Ramsey planner would be a potentially fruitful extension of this model.

Figure 2: Low ($\sigma = 0.1$) and high EIS ($\sigma = 2$)

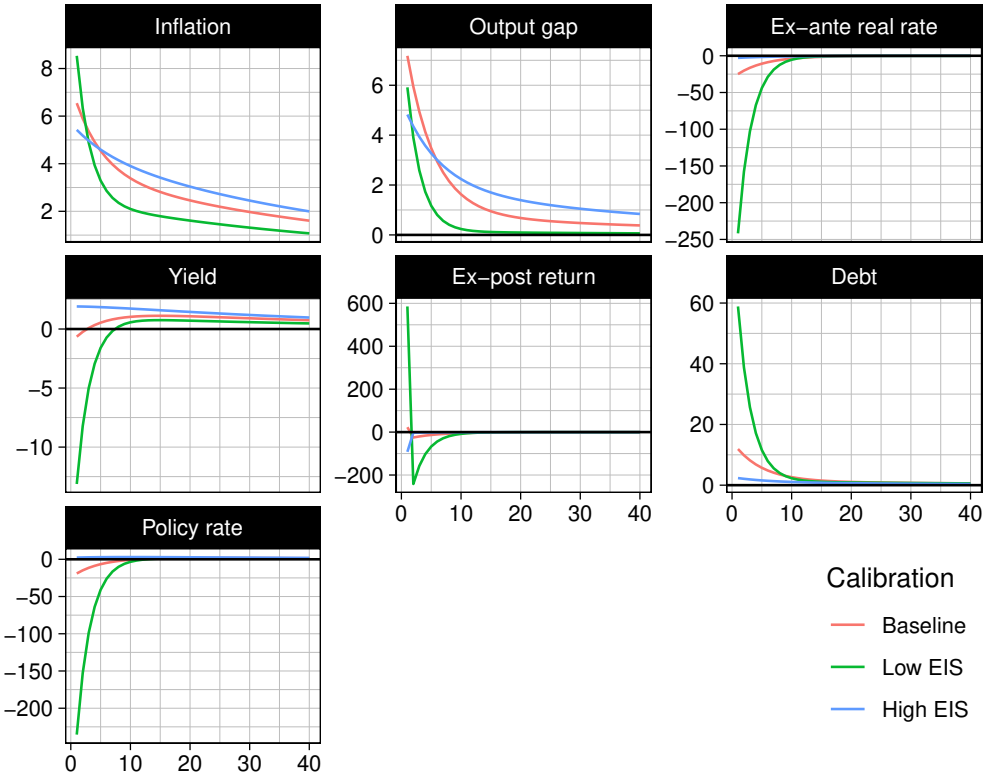
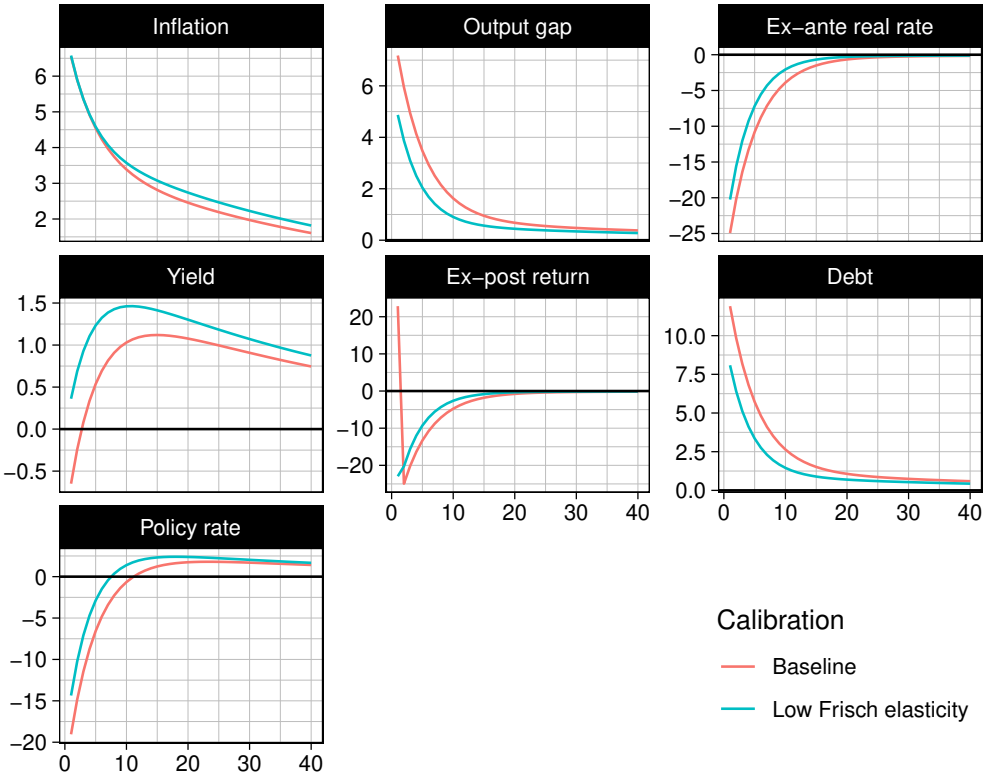


Figure 2 shows how optimal policy reacts to very low ($\sigma = 0.1$) and very high ($\sigma = 2$) elasticity of intertemporal substitution (EIS). With very low EIS, consumption is very insensitive to changes in real interest rates, so the government needs to engineer a massive reduction in nominal interest rates to get the inflation rate to react. The size of the initial change in i_t, which already beggars belief in the baseline calibration, is patently absurd, not to mention the fact that our log-linear approximation to the equilibrium conditions of the economy is likely to break down in the face of such a large deviation from steady state. The scenario with high EIS yields more reasonable results: the planner must now play carefully with real interest rates to avoid large losses coming from inflation and output, because the household is very sensitive to changes in the intertemporal price of consumption. The reduction in debt is now achieved by raising *both* inflation and nominal interest rates, which accomplishes most of the reduction in the real value of the debt stock in the initial period.

Changing the Frisch elasticity of labour supply so that it is consistent with microeconomic evidence does not alter the results. The impulse responses in Figure 3 are very similar to those in Figure 1, with the main difference being that there is no windfall to bondholders in the initial period, because the yield jumps above steady state from the beginning. Intuitively, a lower ψ implies a greater degree of real rigidities in our economy, because the labour supply is less responsive to changes in the wage rate. This enables the planner to obtain the desired level of inflation with a lower output gap, and thus, a smaller reduction in interest rates, so the effect of inflation on bond prices dominates over the interest rate effect.

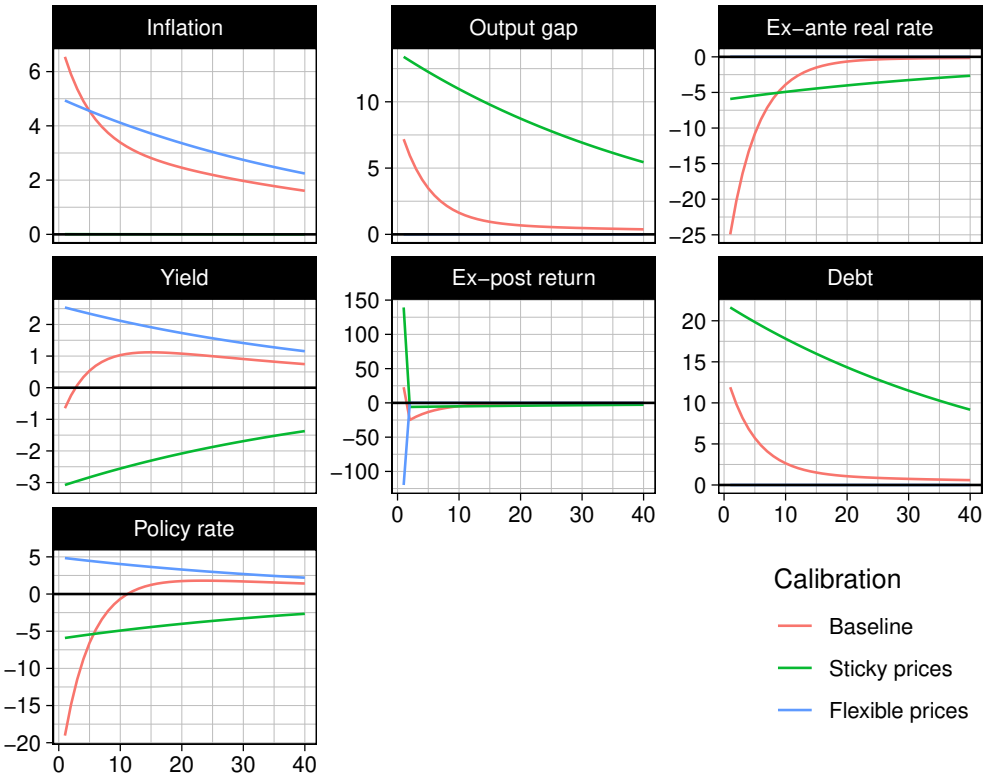
Figure 3: Low Frisch elasticity ($\psi = 0.75$)



Finally, the cases of very sticky and flexible prices (Figure 4) are very instructive about the trade-off the planner is facing. When prices are almost completely fixed, the government cannot inflate away the debt, so it commits to a policy of negative interest rates to reduce the debt stock. This leads to a large and persistent output gap. In contrast, when prices are almost completely flexible, the ex-ante real interest rate is pinned down by the real forces of the economy, and the welfare-maximising policy quickly inflates away the debt. The high inflation rates are almost innocuous for welfare, since the real and nominal sides of the economy become independent of each

other as $\theta \to 0$ (the classical dichotomy).

Figure 4: Very sticky ($\theta = 0.999$) and flexible prices ($\theta = 0.001$)

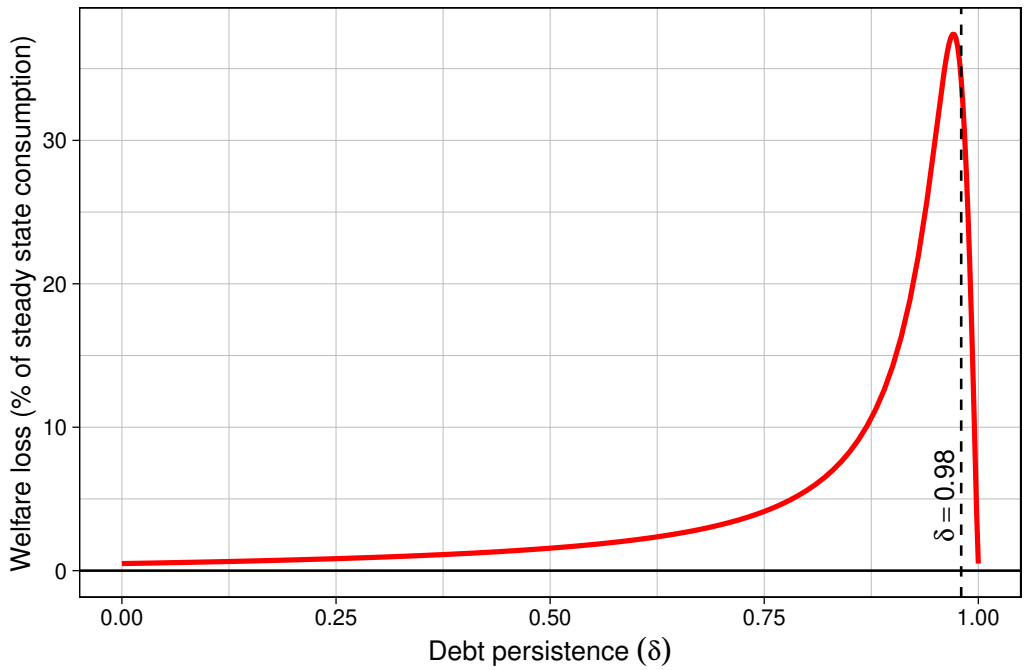


2.4 Optimal debt maturity

Optimal policy in the baseline calibration involves a welfare loss equivalent to 34.23% of steady-state consumption over the first 200 months after a fiscal shock of 10% of the debt stock. This takes the debt persistence parameter, δ, as given, but in a more realistic setting governments would *choose* this parameter optimally, to minimise the welfare loss associated with the fiscal shocks. Instead of modelling this choice explicitly, I use the model calibrated

in the last section and explore numerically what happens to the welfare loss as δ goes from 0 to 1. The results of this exercise can be seen in Figure 5.

Figure 5: Welfare loss after a fiscal shock



Curiously enough, my original calibration ($\delta = 0.98$) is very close to the global maximum of the loss function. The hump shape indicates that losses are minimised at the extremes, when $\delta = 0$ (loss of 0.4919%) and $\delta = 1$ (loss of 0.4907%). Figure 6 shows the impulse responses in these two cases.

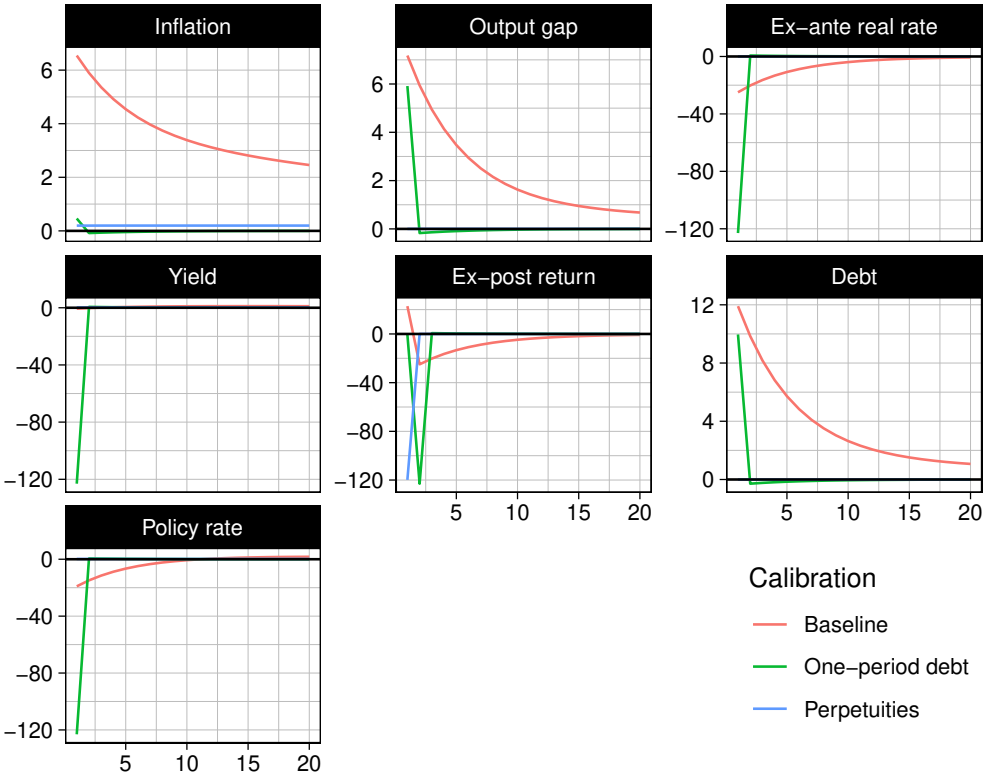
The $\delta = 0$ case corresponds to one-period bonds, which is the usual setup in the basic New Keynesian model. When all of the debt stock needs to be refinanced every period, reducing the debt via the ex-post real return becomes too expensive in terms of the welfare loss. We can see this by looking at equation 5, which collapses to $R_t = j_{t-1} - \pi_t = i_{t-1} - \pi_t$ when $\delta = 0$. That

is to say, to expropriate wealth from current bondholders, the government must increase current inflation one-for-one. In contrast, changes in ex-ante real rates are particularly effective in reducing the debt burden, because all of the stock is refinanced at the new rate. The positive effect on welfare results from the fact that real interest rates only need to be manipulated in the initial period, with the economy returning to steady state very quickly compared to the baseline scenario in section 2.3.

The $\delta = 1$ case corresponds to perpetuities. The Ramsey policy in this case involves mild but persistent inflation, running at 0.2% per annum indefinitely, and a similar path for the policy rate and the yield. Intuitively, manipulating ex-ante interest rates by reducing the policy rate becomes prohibitively expensive for the government, because the price of perpetuities is very sensible to interest rates (recall that the Macaulay duration can be interpreted as the elasticity of the bond price with respect to the yield, and duration is increasing in δ). Hence, the planner would rather *raise* the yield to immediately reduce the debt burden. The welfare loss is very low despite the persistence in the inflationary process, because the price level is growing very slowly along a constant path and firms are able to anticipate future increases in marginal costs when setting prices, making the output gap negligible.

A final question we may ask ourselves is: why is the welfare loss particularly large for values of δ between 0.9 and 0.99? A possible interpretation is that there are very few bonds being refinanced every period, so reducing the debt stock using negative ex-ante real interest rates requires a long period of

Figure 6: One-period bonds ($\delta = 0$) and perpetuities ($\delta = 1$)



distortionary inflation, and the high sensitivity of the bond price to interest rates means that the initial debt burden is amplified. Doing most of the adjustment in the initial period via the ex-post return on existing debt is not an attractive alternative either, because raising the bond yield makes issuing new debt more expensive, and eventually the full debt stock will be replaced by debt issued with the higher interest rates.

Given that the term structure of government debt in most developed economies corresponds to a value of δ between 0.9 and 0.99, the model suggests that financial repression is very costly. Governments would do well in

considering alternative means of responding to fiscal shocks, such as increases in taxes or reductions in government spending.

3 Financial repression in a New Keynesian model with capital

3.1 Setup

3.1.1 Households

A representative household chooses consumption, labour supply, bond holdings and capital to maximise the discounted present value of CRRA utility, additively separable in consumption and labour:

$$\max_{\{C_{t+h},N_{t+h},K_{t+h},B_{t+h}\}} \mathbb{E}_t \sum_{h=0}^{\infty} \beta^h \frac{C_{t+h}^{1-\frac{1}{\sigma}} - 1}{1-\frac{1}{\sigma}} + \frac{N_{t+h}^{1+\frac{1}{\psi}}}{1+\frac{1}{\psi}}$$

$$\begin{aligned} \text{s.t. } C_{t+h} + K_{t+h} + \frac{Q_{t+h}B_{t+h}}{P_{t+h}} \leq & \frac{W_{t+h}N_{t+h}}{P_{t+h}} + (1 + r_t^k - \tau)K_{t+h-1} \\ & + \frac{(1+\delta Q_{t+h})B_{t+h-1}}{P_{t+h}} + D_{t+h} - T_{t+h} \end{aligned}$$

Notice that K_t and B_t denote *end-of-period* capital and bond holdings, respectively, to conform to Dynare's timing convention (all t-dated variables are chosen at time t). The FOC of the household's problem, after removing the Lagrange multiplier, can be summarised by a labour supply condition and two Euler equations (one for bonds and one for capital):

$$N_t^{\frac{1}{\psi}} = w_t C_t^{-\frac{1}{\sigma}} \tag{16}$$

$$\frac{Q_t}{P_t} C_t^{-\frac{1}{\sigma}} = \beta \mathbb{E}_t \left[C_{t+1}^{-\frac{1}{\sigma}} \frac{(1 + \delta Q_{t+1})}{P_{t+1}} \right] \tag{17}$$

$$C_t^{-\frac{1}{\sigma}} = \beta \mathbb{E}_t \left[C_{t+1}^{-\frac{1}{\sigma}} (1 + r_t^k - \tau) \right] \tag{18}$$

Where $w_t := \frac{W_t}{P_t}$ is the real wage.

Consumption C_t and investment $I_t = K_t - (1 - \tau)K_{t-1}$ are standard CES aggregators of a continuum of goods indexed by $i \in [0, 1]$. For simplicity, assume that all goods can be bought for either consumption or investment at the same price. Also, both aggregators are the same: $C_t := \left(\int_0^1 c_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}$ and $I_t := \left(\int_0^1 I_t(i)^{\frac{\varepsilon-1}{\varepsilon}} di \right)^{\frac{\varepsilon}{\varepsilon-1}}$.

Households decide which individuals goods to buy by minimising the cost of obtaining a given level of aggregate consumption and investment. The resulting demand schedules have the usual constant-elasticity form:

$$I_t(i) = \left(\frac{p_t(i)}{P_t} \right)^{-\varepsilon} I_t \tag{19}$$

$$c_t(i) = \left(\frac{p_t(i)}{P_t} \right)^{-\varepsilon} C_t \tag{20}$$

Finally, the price index P_t is defined as $P_t = \left(\int_0^1 p_t(i)^{1-\varepsilon} di \right)^{\frac{1}{1-\varepsilon}}$.

3.1.2 Cost minimisation

There is a continuum of monopolists indexed by i, each of which manufactures a unique variety of consumption and investment goods. Each firm has an

identical Cobb-Douglas production function:

$$Y_t(i) = A_t K_{t-1}(i)^{\alpha} N_t(i)^{1-\alpha} \tag{21}$$

where $K_{t-1}(i)$ and $N_t(i)$ are the capital and labour inputs used by firm i at time t. Input markets are assumed to be perfectly competitive, with all firms paying a real rental rate r_t^k and a real wage w_t to hire capital and labour from households, respectively. Notice the subscript on capital: firms can only hire capital out of the aggregate stock K_{t-1}, which is available at the beggining of period t.

Since firms do not make investment decisions, the selection of the optimal input mix is a static problem that can be solved via standard per-period cost-minimisation. The cost-minimising capital-labour ratio is:

$$\frac{K_{t-1}(i)}{N_t(i)} = \frac{\alpha}{1-\alpha} \frac{w_t}{r_t^k} \tag{22}$$

Finally, minimised real cost is $X_t(Y_t(i)) = x_t Y_t(i)$, where $\eta := \alpha^{-\alpha}(1-\alpha)^{-(1-\alpha)}$ and x_t represents marginal cost:

$$x_t := \eta (r_t^k)^{\alpha} w_t^{1-\alpha} \frac{1}{A_t} \tag{23}$$

Notice that marginal costs are constant (a consequence of the production technology exhibiting CRS) and do not depend on i.

3.1.3 Aggregation

Goods markets clear at the firm level, $c_t(i) + I_t(i) = Y_t(i)$, and at the aggregate level, $C_t + I_t = Y_t$, where Y_t is defined as the same CES aggregator of firm-level output $Y_t(i)$.[2] This means that total demand for each good i also follows the familiar constant-elasticity form:

$$Y_t(i) = \left(\frac{p_t(i)}{P_t}\right)^{-\varepsilon} Y_t \tag{24}$$

Factor markets must also clear. That is to say, $K_{t-1} = \int_0^1 K_{t-1}(i)di$ and $N_t = \int_0^1 N_t(i)di$. Since the firm-level capital-labour ratio defined in equation 22 is constant across firms, it is equal to the aggregate capital-labour ratio. Aggregating the individual production functions then gives us the aggregate production function, which is Cobb-Douglas with a productivity wedge represented by the price dispersion index $\Delta_t := \int_0^1 \left(\frac{p_t(i)}{P_t}\right)^{-\varepsilon} di$:

$$Y_t = \frac{A_t}{\Delta_t} K_{t-1}^{\alpha} N_t^{1-\alpha} \tag{25}$$

Now, since the aggregate capital stock K_{t-1} is predetermined, then for any level of aggregate output, the required labour input is $N_t = \left(\frac{\Delta_t Y_t}{A_t K_{t-1}^{\alpha}}\right)^{\frac{1}{1-\alpha}}$. Clearing in the labour market requires that the wage adjusts to satisfy this as well as the labour supply condition (equation 16). This gives us:

$$w_t = \frac{\Delta_t^{\frac{1}{\psi(1-\alpha)}} Y_t^{\frac{1}{\psi(1-\alpha)}+\frac{1}{\sigma}}}{A_t^{\frac{1}{\psi(1-\alpha)}} K_{t-1}^{\frac{\alpha}{\psi(1-\alpha)}}} \left(\frac{C_t}{Y_t}\right)^{\frac{1}{\sigma}} \tag{26}$$

[2] To see this, plug the consumption and investment demand schedules into $Y_t(i) = c_t(i) + I_t(i)$ and then aggregate over i using the CES aggregator.

Finally, the real rental rate adjusts so that the capital-labour ratio stays constant:

$$r_t^k = \frac{\alpha}{1-\alpha} w_t \left(\frac{\Delta_t Y_t}{A_t K_{t-1}} \right)^{\frac{1}{1-\alpha}} \tag{27}$$

3.1.4 Price setting

As in the baseline model, I assume time-dependent Calvo (1983) pricing, where a fraction θ of randomly selected firms cannot change their prices at any time period t. Each firm solves the following reset price problem:

$$\max_{P_t^*} \mathbb{E}_t \sum_{h=0}^{\infty} \theta^h q_{t+h|t} Y_{t+h} \left[\left(\frac{P_t^*}{P_{t+h}} \right)^{1-\varepsilon} - (1-\varsigma) x_{t+h} \left(\frac{P_t^*}{P_{t+h}} \right)^{-\varepsilon} \right]$$

Where $q_{t+h|t} := \beta^h \frac{C_{t+h}^{-\frac{1}{\sigma}}}{C_t^{-\frac{1}{\sigma}}}$ is the stochastic discount factor, x_t is marginal cost and ς is a marginal cost subsidy (to be discussed shortly). The first order condition for a maximum is:

$$\mathbb{E}_t \sum_{h=0}^{\infty} \theta^h q_{t+h|t} Y_{t+h} \left(\frac{P_t^*}{P_{t+h}} \right)^{-\varepsilon} \left[\frac{P_t^*}{P_{t+h}} - (1-\varsigma) \frac{\varepsilon}{\varepsilon - 1} x_{t+h} \right] = 0 \tag{28}$$

Notice that in the special case with perfectly flexible prices ($\theta = 0$), $P_t = P_t^*$, so the reset price equation collapses to the familiar monopoly pricing rule $x_t = (1-\varsigma)\frac{\varepsilon-1}{\varepsilon}$. To ensure that the flexible-price equilibrium outcome

(and thus, the zero-inflation steady state around which I log-linearize the system) is efficient, set the cost subsidy $\varsigma = \frac{1}{\varepsilon}$, so that it exactly offsets the inefficiency coming from monopolistic competition. This ensures that $x_t = 1$ in steady state, allowing us to focus on the inefficiencies caused by nominal rigidities.[3]

3.1.5 Government

The government sector is identical to the one in the baseline model. That is to say, the public sector issues bonds B_t, each of which entitles the bondholder to a stream of payments $\{1, \delta, \delta^2, \delta^3, \dots\}$. Revenues come from lump sum taxation (net of transfers) T_t and the model abstracts from government expenditures, so the government's budget identity is:

$$T_t + \frac{Q_t B_t}{P_t} = \frac{(1 - \delta Q_t) B_{t-1}}{P_t}$$

There are fiscal shocks affecting revenues T_t around its steady state value, such that $\phi_t = \frac{\bar{T} - T_t}{\bar{D}}$, where $\bar{D} := \frac{\bar{Q}\bar{B}}{\bar{P}}$ is the steady state value of the real debt stock.

[3]This is by no means an innocuous assumption, but it greatly simplifies the analysis. The presence of imperfectly competitive product markets in a neoclassical growth model makes the steady-state level of capital per unit of labour inefficiently low, because market power in product markets introduces a wedge between the rental rate and the marginal product of capital (Galí, 1996). The subsidy gets rid of the wedge, ensuring that the economy's steady state is efficient.

3.2 Log-linearised model

Log-linearising the equilibrium conditions around a steady state with zero inflation yields the following system of linear difference equations:

$$\pi_t = \beta \mathbb{E}_t(\pi_{t+1}) + \xi \hat{x}_t \tag{29}$$

$$\hat{x}_t = \frac{1}{\gamma_0}\hat{r}_t^k + \hat{k}_{t-1} - \hat{y}_t \tag{30}$$

$$c_t = \mathbb{E}_t(c_{t+1}) - \sigma r_t \tag{31}$$

$$\hat{y}_t = (1-\bar{s})c_t + \bar{s}\hat{I}_t \tag{32}$$

$$\hat{k}_t = (1-\tau)\hat{k}_{t-1} + \tau \hat{I}_t \tag{33}$$

$$\hat{r}_t^k = \gamma_0 \left[\gamma_1 \hat{y}_t - \gamma_2 \hat{k}_{t-1} + \frac{1}{\sigma}(c_t - \hat{y}_t)\right] \tag{34}$$

$$r_t = \mathbb{E}_t(R_{t+1}) \tag{35}$$

$$r_t = \mathbb{E}_t(\hat{r}_{t+1}^k) \tag{36}$$

where the following parameters are defined as:

$\gamma_0 := 1-\beta(1-\tau)$, $\gamma_1 := \frac{1}{\sigma} + \left(1+\frac{1}{\psi}\right)\frac{1}{1-\alpha}$, $\gamma_2 := \left(1+\frac{\alpha}{\psi}\right)\frac{1}{1-\alpha}$, $\bar{s} := \frac{\alpha\beta\tau}{1-\beta+\beta\tau}$ and $\xi := \frac{(1-\beta\theta)(1-\theta)}{\theta}$

See Appendix B for details on the derivation. These conditions, together with the public debt block from the baseline model (equations 4 through 6), form a system of 11 equations and 12 endogenous variables, which means there is one degree of freedom left to implement a monetary policy rule. To obtain this, I assume the planner naïvely minimises the *same* loss function as

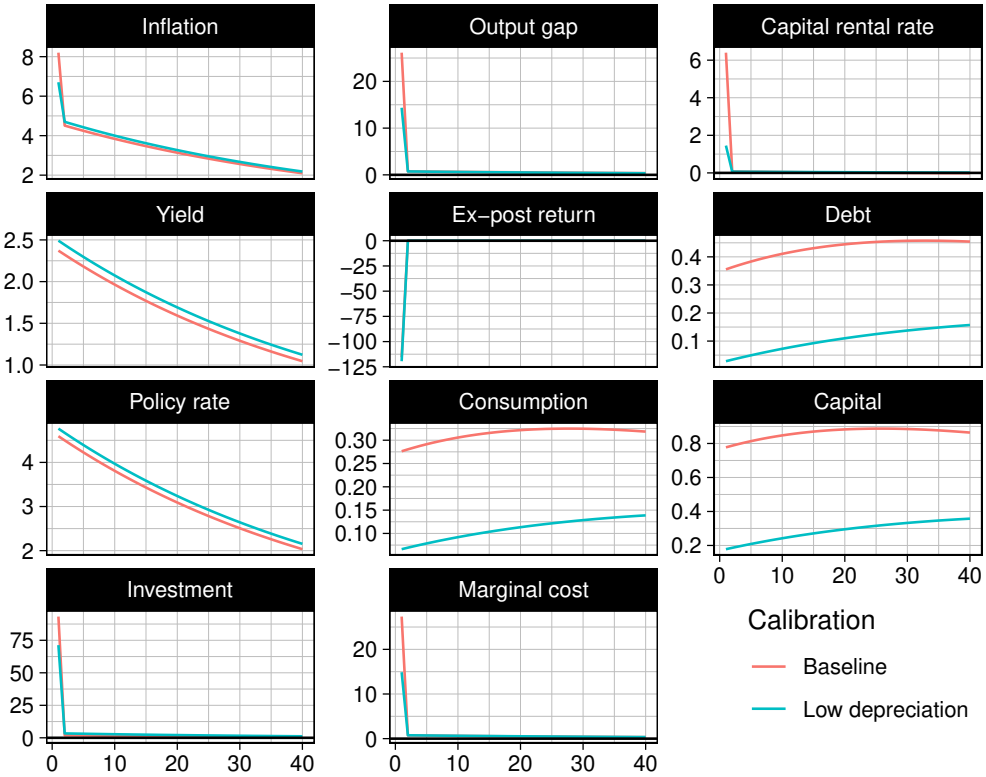
in section 2.2, not realising that it no longer has the same welfare implications as in the economy without capital (see section 3.4 for a discussion of this issue).

3.3 Calibration and simulation

Compared to the baseline economy, only one additional parameter needs to be calibrated: the depreciation rate of capital. Following Smets and Wouters (2007), I set the annual depreciation rate equal to 10%, which implies a monthly depreciation of approximately $\tau \approx 0.00834$. This has implications for the steady state saving rate in our model, which is equal to 27.82% with our baseline calibration. Since this seems a little high for advanced economies, I also experiment with a lower depreciation rate of $\tau = 0.0025$, which is equivalent to an annual depreciation rate of 3% and leads to a steady-state saving rate equal to 20.07%. The resulting impulse responses are depicted in Figure 7.

The introduction of capital into the model has a subtle effect on financial repression. Since the expected return on government bonds is now linked to the expected return on capital via arbitrage, this limits the ability of the government to implement financial repression via manipulation of the ex-ante real rate. For instance, with perfect foresight, a rise in future inflation pushes down the expected real return on government debt (equation 5), but with sticky prices, a positive output gap will translate into a *higher* rental rate of capital. The Fisher equation (35) implies that the expected return to both assets in the economy must be equal in equilibrium, so the initial effect

Figure 7: Capital baseline and low depreciation ($\tau = 0.0025$)



on the ex-ante real rate will be dampened. Intuitively, real assets become more attractive than nominal assets, which increases investment demand and reduces demand for government bonds, lowering their price and raising their expected return.

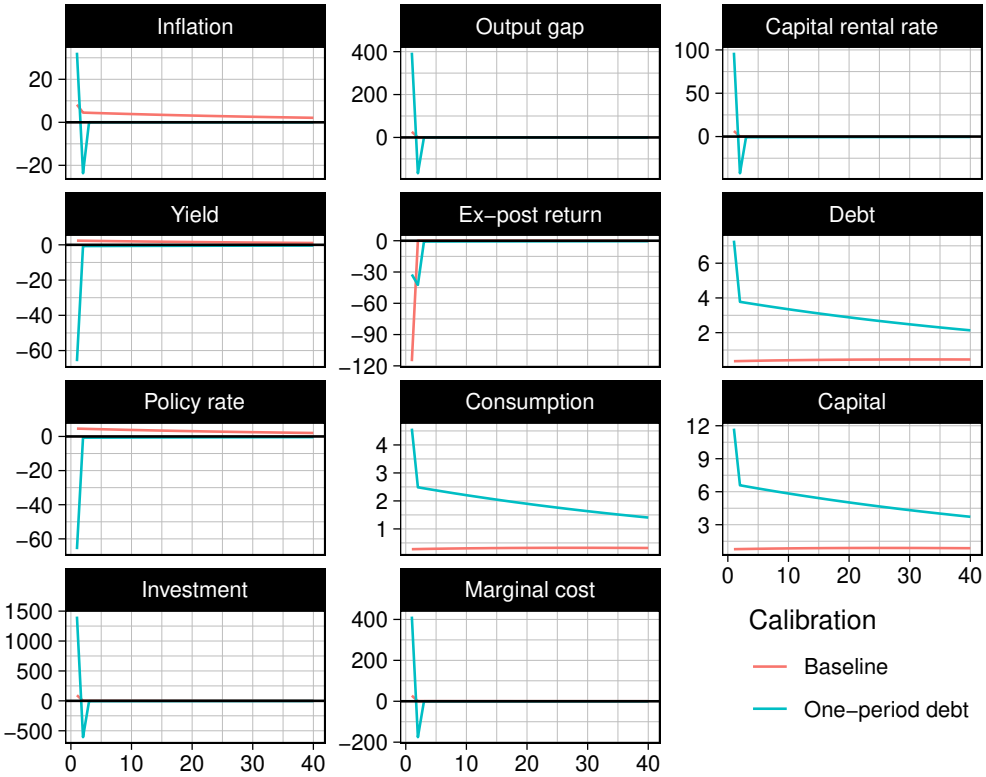
We can see the weakening of the ex-ante channel in the IRFs. In contrast with the results in Figure 1, now most of the adjustment in the debt stock comes from the initial inflationary shock and rise in the yield, which reduce the ex-post return for the original bondholders. The rest of the reduction in the debt stock is achieved by letting real interest rates become negative *after*

the initial rise in the capital stock (another manifestation of the link between the return on both assets in the economy). Changing the depreciation rate does not alter this fundamental result, although it does affect the size of the changes in the output gap and in the capital stock. A possible interpretation is that with a lower depreciation rate, increases in the capital stock take much longer to be reverted, so the government reduces the size of the initial jump in the output gap to avoid a more protracted deviation from steady state.

The relative weakness of the ex-ante channel is evident when we focus on the one-period-debt case (Figure 8). Recall that, in the model without capital, the optimal Ramsey policy in the $\delta = 0$ case involved a large reduction in the ex-ante real rate in the initial period, taking advantage of the fact that all of the debt stock had to be refinanced at the new rate. This was achieved with relatively mild changes in the inflation rate, thus leading to a very low welfare loss. With endogenous capital accumulation, the required changes in inflation and the output gap are two orders of magnitude greater, with the government engineering a massive swing in inflation from 32.37% in the initial period to -23.69% in the next period. Such a large disinflation pushes the expected rental rate on capital to deeply negative territory, enabling the government to reduce the debt thanks to the expected negative return on nominal bonds.

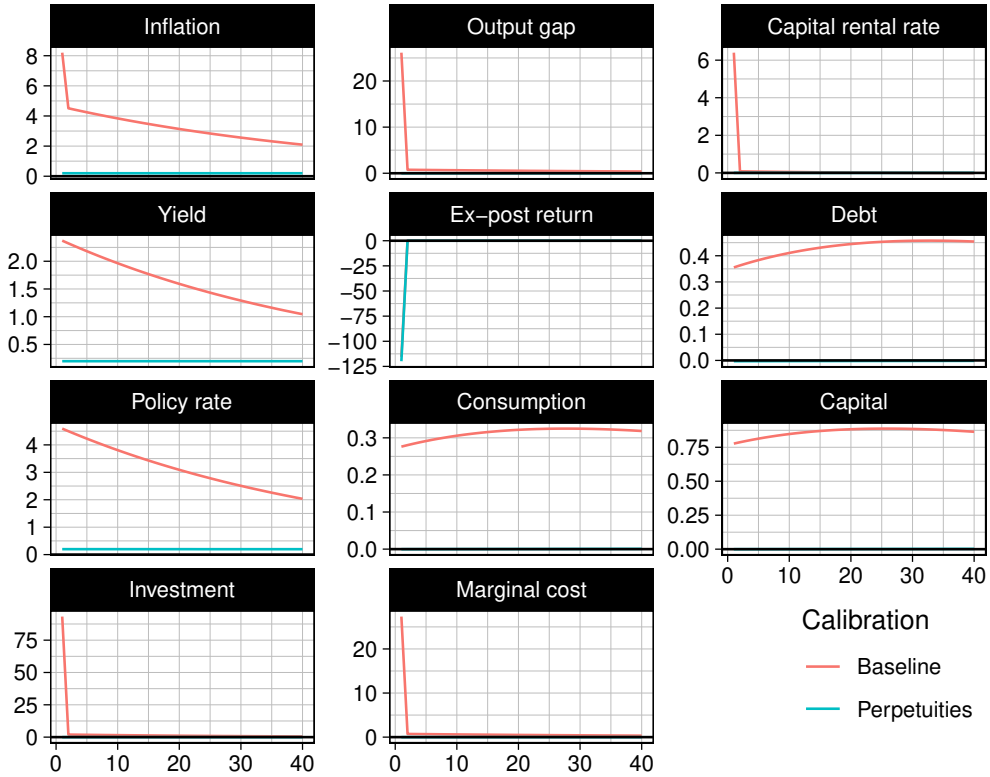
In contrast, policy in the perpetuity case (Figure 9) closely matches the results in section 2.3. The fiscal adjustment happens almost exclusively via the ex-post return channel, with the government engineering a gradual ex-

Figure 8: Capital and one-period bonds ($\delta = 0$)



pansion in the price level and the real variables of the economy experiencing only very small changes (on the order of 0.001% of steady-state values). This agrees with the view that capital accumulation leaves the potency of the ex-post return channel unaffected. This makes intuitive sense, since the arbitrage between nominal and real assets only holds in *expectation*, and the rates of return may diverge substantially in the initial period.

Figure 9: Capital and perpetuities ($\delta = 1$)



3.4 Welfare

When capital is introduced into the New Keynesian model, the loss function presented in equation 8 is no longer a valid approximation to the loss in utility compared to the steady state. Thus, government policy as simulated in section 3.3 should *not* be interpreted as the Ramsey or Pareto-optimal policy choice. That does not mean that the exercise is useless: according to Woodford (2003, p. 401), loss functions of this form were popular in the literature on monetary policy evaluation before the formal welfare implications were derived. So long as governments and central banks attempt to

minimize some combination of deviations of output (or unemployment) and inflation from target, the policy obtained by minimising 8 could represent a reasonable approximation to their behaviour.

Developing an alternative welfare-based loss function for the New Keynesian model with capital is outside the scope of this essay, but a few brief comments about the welfare implications of financial repression can be ventured here.[4] The main difference with the basic New Keynesian model is that the relationship between the output gap and welfare is more complicated, because it depends on the capital stock and the saving rate. Namely, a positive output gap may now *increase* welfare if it is due to a higher capital stock, because labour supply need not have increased and may even have gone down. Similarly, for a given output gap, a lower saving rate (which is equivalent to lower investment in a closed economy) will increase the flow welfare of the household by raising the fraction of output devoted to consumption.

These two effects dampen the welfare losses stemming from the initial increases in inflation and the output gap in the baseline calibration of Figure 7. Not much can be said about the magnitude of the overall change in welfare caused by the fiscal shock, but the sign is surely negative. Recall that at the steady state, additional increases in the capital stock are suboptimal for the household, because the marginal cost of additional investment is larger than the marginal value of additional capital (which in turn comes from higher future consumption). Thus, tricking the household into investing more

[4]Note that, as discussed by Woodford (2005), the introduction of endogenous capital accumulation raises thorny issues about the definition of the output gap and the natural level of output. For our purposes, the output gap refers to the difference between (the log of) current and steady state output.

by artificially raising the expected rental rate on capital, without actually changing its marginal product, can never increase welfare.

Lastly, it is likely that the result obtained in section 2.4 breaks down when capital is introduced into the model. In particular, as observed in Figure 8, financial repression policy in the $\delta = 0$ case is now much more costly. The large output gap in the initial period (395.18% of steady state output) is entirely negative for welfare, since the available capital stock is still stuck at steady state and most of the additional output represents higher investment, not consumption. It is true that consumption stays above the steady-state level for longer, while the output gap returns quickly to 0, which suggests a positive effect on welfare due to a lower saving rate. However, the size of the initial output gap is likely to overwhelm the consumption effect, generating large welfare losses. In contrast, financial repression in the $\delta = 1$ case in Figure 9 exhibits negligible effects on real outcomes, just like in the basic model. Hence, very long debt durations are probably preferable to short-term debt when capital accumulation is taken into account.

4 Conclusion

The analysis in the preceding sections has shown that financial repression has substantial costs for the more realistic calibrations of the simple New Keynesian model. To accommodate fiscal shocks, the planner induces persistent inflation, generating large welfare losses stemming from relative price distortions and positive output gaps. This is because, for more realistic gov-

ernment debt durations, the two channels through which financial repression works (sudden, ex-post expropriation of the returns of current bondholders, and anticipated negative ex-ante real rates) are relatively weak and require prolonged departures from the first-best allocation. When capital accumulation is taken into account, this weakness is amplified by the fact that the real return on government bonds is linked to the return on real assets (i.e., capital) via arbitrage, and reductions on the real interest rate on bonds can only be achieved if one is willing to induce a reduction in the capital stock of the economy.

This is not enough to judge the merits of financial repression policy, since it may well be that the alternatives are worse. The framework used above abstracts from several crucial policy levers that governments may employ: lower expenditures, higher taxes, growth-enhancing structural reforms and even outright default (Reinhart et al., 2015). Some of these are potentially very costly; for instance, increases in taxes may worsen distortions, reductions in public expenditure or transfers may have perverse effects during a recession and default may substantially limit the choices of future governments. A complete assessment of the optimal response to fiscal shocks would require us to weigh the costs and benefits of all these policy options in a unified framework. Building on the analysis carried out in this essay by introducing some of these aspects into the New Keynesian model thus appears to be a very promising avenue for further research.

References

Barkai, S. (2020). Declining labor and capital shares. *The Journal of Finance (New York)*, *75*(5), 2421–2463.

Basu, S. (2019). Are price-cost markups rising in the United States? a discussion of the evidence. *The Journal of Economic Perspectives*, *33*(3), 3–22.

Calvo, G. A. (1983). Staggered prices in a utility-maximizing framework. *Journal of Monetary Economics*, *12*(3), 383–398.

Chetty, R., Guren, A., Manoli, D., & Weber, A. (2011). Are micro and macro labor supply elasticities consistent? A review of evidence on the intensive and extensive margins. *The American Economic Review*, *101*(3), 471–475.

Galí, J. (1996). Multiple equilibria in a growth model with monopolistic competition. *Economic Theory*, *8*(2), 251–266.

Galí, J. (2015). *Monetary policy, inflation, and the business cycle: An introduction to the New Keynesian framework and its applications* (Second edition.).

Havranek, T., Horvath, R., Irsova, Z., & Rusnak, M. (2015). Cross-country heterogeneity in intertemporal substitution. *Journal of International Economics*, *96*(1), 100–118. https://doi.org/https://doi.org/10.1016/j.jinteco.2015.01.012

Nakamura, E., & Steinsson, J. (2008). Five facts about prices: A reevaluation of menu cost models. *The Quarterly Journal of Economics*, *123*(4), 1415–1464.

OECD. (2022). *OECD Sovereign Borrowing Outlook 2022*. https://doi.org/https://doi.org/https://doi.org/10.1787/b2d85ea7-en

Reinhart, C. M., Reinhart, V., & Rogoff, K. (2015). Dealing with debt [37th Annual NBER International Seminar on Macroeconomics]. *Journal of International Economics*, *96*, S43–S55. https://doi.org/https://doi.org/10.1016/j.jinteco.2014.11.001

Smets, F., & Wouters, R. (2007). Shocks and frictions in US business cycles: A Bayesian DSGE approach. *The American Economic Review*, *97*(3), 586–606.

Woodford, M. (2003). *Interest and prices: Foundations of a theory of monetary policy*. Princeton University Press.

Woodford, M. (2005). Firm-specific capital and the New Keynesian Phillips Curve. *International Journal of Central Banking*, *1*(2). https://ideas.repec.org/a/ijc/ijcjou/y2005q3a1.html

Appendix A: Derivation of the analytical solution to the Ramsey problem

The Ramsey planner minimises equation 8 subject to equations 1 through 6. The set of FOC for a minimum comprises the constraints and the following conditions:

$$\frac{\varepsilon}{\nu\xi}\pi_t - \lambda_t^A + \lambda_{t-1}^A - \lambda_t^E = 0 \tag{37}$$

$$\gamma y_t + \kappa\lambda_t^A - \lambda_t^B + \beta^{-1}\lambda_{t-1}^B = 0 \tag{38}$$

$$-\sigma\lambda_t^B + \lambda_t^C = 0 \tag{39}$$

$$\frac{\beta}{1-\beta\delta}\mathbb{E}_t(\lambda_{t+1}^E) - \frac{\beta\delta}{1-\beta\delta}\lambda_t^E - \lambda_t^D + \delta\lambda_{t-1}^D = 0 \tag{40}$$

$$\lambda_t^D(1-\beta\delta) = 0 \tag{41}$$

$$-\lambda_t^F + \lambda_t^E - \beta^{-1}\lambda_{t-1}^C = 0 \tag{42}$$

$$\mathbb{E}_t(\lambda_{t+1}^F) - \lambda_t^F = 0 \tag{43}$$

Where λ_t^A is defined to be the Lagrange multiplier associated with equation 1, λ_t^B to be the Lagrange multiplier associated with equation 2, and so on until λ_t^F.

First, notice that since $\beta < 1$ and $\delta \in [0, 1]$, equation 41 implies that $\lambda_t^D = 0$ for all t. Thus, equation 40 collapses to:

$$\mathbb{E}_t(\lambda_{t+1}^E) = \delta\lambda_t^E \tag{44}$$

It is also possible to use 39 to get rid of the third multiplier, which satisfies $\lambda_t^C = \sigma\lambda_t^B$.

Next, conjecture that the multipliers associated with the last two constraints follow:

$$\lambda_t^E = \delta\lambda_{t-1}^E + \chi\phi_t \tag{45}$$

$$\lambda_t^F = \lambda_{t-1}^F + \chi\phi_t \tag{46}$$

Where χ is a common parameter, to be found later. Clearly, these solutions satisfy equations 44 and 43 since the fiscal shock ϕ_t is an innovation.

Using equation 42 and the conjecture, λ_t^B can be expressed as a linear combination of the λ_t^E and λ_t^F random variables:

$$\lambda_t^B = \frac{\beta}{\sigma}(\lambda_{t+1}^F - \lambda_{t+1}^E) = \frac{\beta}{\sigma}(\lambda_t^F - \delta\lambda_t^E)$$

Then, substituting out λ_t^B and solving for inflation and the output gap in equations 37 and 38 yields:

$$\pi_t = \frac{\nu\xi}{\varepsilon}\left(\lambda_t^A - \lambda_{t-1}^A + \lambda_t^E\right) \tag{47}$$

$$y_t = \gamma^{-1}\left(\frac{1-\beta\delta}{\sigma}\lambda_t^E - \frac{1-\beta}{\sigma}\lambda_t^F - \kappa\lambda_t^A\right) \tag{48}$$

Substituting out inflation and the output gap in the Phillips curve (eq. 1), then after simplification the following second-order difference equation for λ_t^A emerges:

$$(1+\beta+\varepsilon\kappa)\lambda_t^A - \beta\mathbb{E}_t(\lambda_{t+1}^A) - \lambda_{t-1}^A = \left[\frac{\varepsilon(1-\beta\delta)}{\sigma} - (1-\delta)\right]\lambda_t^E - \frac{\varepsilon(1-\beta)}{\sigma}\lambda_t^F$$

Guessing that a stationary solution of the form

$$\lambda_t^A = a_1\lambda_{t-1}^A + a_2\lambda_t^E + a_3\lambda_t^F \tag{49}$$

with $a_1 < 1$, exists, the method of matching coefficients yields the following parameter values:

$$a_1 = \frac{1+\beta+\varepsilon\kappa - \sqrt{(1+\beta+\varepsilon\kappa)^2 - 4\beta}}{2\beta} \tag{50}$$

$$a_2 = \frac{\varepsilon(1-\beta\delta) - \sigma(1-\delta)}{(1+\beta+\varepsilon\kappa-\beta\delta-\beta a_1)\sigma} \tag{51}$$

$$a_3 = -\frac{\varepsilon(1-\beta)}{(1+\varepsilon\kappa-\beta a_1)\sigma} \tag{52}$$

It can be verified that $a_1 < 1$, validating the initial guess. Next, substitute out inflation and the output gap from the IS curve (eq. 2) to obtain an equation for the ex-ante real interest rate in terms of the multipliers:

$$r_t = \frac{(1-\beta\delta)(\delta-1)}{\gamma\sigma^2}\lambda_t^E - \frac{\kappa}{\gamma\sigma}[\mathbb{E}_t(\lambda_{t+1}^A) - \lambda_t^A] \tag{53}$$

Then, using equations 3 and 5 to substitute out the ex-ante real rate, the following second order equation for j_t can be obtained:

$$\frac{\sigma\gamma}{1-\beta\delta}\left[j_t - \beta\delta\mathbb{E}_t(j_{t+1})\right] = \left[\frac{\sigma\kappa\delta}{\varepsilon} - \frac{(1-\beta\delta)(1-\delta)}{\sigma} + \kappa\left(1-\frac{\sigma}{\varepsilon}\right)(1-\delta)\,a_2\right]\lambda_t^E \\ - \kappa\left(1-\frac{\sigma}{\varepsilon}\right)a_1\left(\lambda_t^A - \lambda_{t-1}^A\right)$$

Again, guess that the solution for the yield is

$$j_t = b_1\lambda_t^E + b_2(\lambda_t^A - \lambda_{t-1}^A) \tag{54}$$

Matching coefficients results in:

$$b_1 = \frac{(1-\beta\delta)}{(1-\beta\delta^2)\sigma\gamma}\left[\frac{\sigma\kappa\delta}{\varepsilon} - \frac{(1-\beta\delta)(1-\delta)}{\sigma} + \kappa(1-\frac{\sigma}{\varepsilon})(1-\delta)a_2\right] \\ - \frac{\beta\delta(1-\delta)}{(1-\beta\delta^2)}b_2a_2 \tag{55}$$

$$b_2 = -\frac{(1-\beta\delta)\kappa}{(1-\beta\delta a_1)\sigma\gamma}(1-\frac{\sigma}{\varepsilon})a_1 \tag{56}$$

Finally, consider the government's intertemporal budget constraint (eq. 7, which assumes the economy starts from steady state at time t):

$$\beta^{-1}R_t + \sum_{h=0}^{\infty}\beta^h\mathbb{E}_t(r_{t+h}) = -\phi_t$$

Recall that, in equilibrium, R_t and r_t are functions of j_t and π_t (at various leads and lags). It is then possible to express R_t and $\mathbb{E}_t(r_{t+h})$ as functions only of λ_t^E and λ_t^F by using 47, 49, and 54 to substitute out the remaining variables. Imposing the guesses from equations 11 and 12, then both sides of the budget constraint are functions only of the shock at time t. In particular, $\beta^{-1}R_t$ can be shown to be:

$$\beta^{-1}R_t = \left[-\frac{(1+a_1+a_3)\nu\xi}{\beta\varepsilon} - \frac{\delta(b_1+b_2(a_2+a_3))}{1-\beta\delta}\right]\chi\phi_t = (-\chi_1-\chi_2)\chi\phi_t \tag{57}$$

The expression for the infinite sum over $\mathbb{E}_t(r_{t+h})$ is trickier to obtain. Starting from equation 53 and getting rid of the expectation on the right-hand side:

$$\begin{aligned} r_{t+h} &= \left[-\frac{(1-\beta\delta)(1-\delta)}{\sigma^2\gamma} + \frac{\kappa(1-\delta)a_2}{\sigma\gamma} \right] \lambda^E_{t+h} - \frac{\kappa a_1}{\sigma\gamma}(\lambda^A_{t+h} - \lambda^A_{t+h-1}) \\ &= -(1-\beta\delta)\chi_3\lambda^E_{t+h} - \frac{\kappa a_1}{\sigma\gamma}(\lambda^A_{t+h} - \lambda^A_{t+h-1}) \end{aligned}$$

This means the infinite sum can be split as follows:

$$\sum_{h=0}^{\infty} \beta^h \mathbb{E}_t(r_{t+h}) = -(1-\beta\delta)\chi_3 \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t(\lambda^E_{t+h}) - \frac{\kappa a_1}{\sigma\gamma} \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t(\lambda^A_{t+h} - \lambda^A_{t+h-1})$$

Since $\mathbb{E}_t(\lambda^E_{t+h}) = \delta^h \lambda^E_t = \delta^h \chi \phi_t$, the first summation can be simplified to $-(1-\beta\delta)\chi_3 \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t(\lambda^E_{t+h}) = -\chi_3\chi\phi_t$.

For the second summation, the following result (easily derived from eq. 49) is useful:

$$\mathbb{E}_t(\lambda^A_{t+h} - \lambda^A_{t+h-1}) = -a_2(1-\delta)\delta^{h-1}\frac{1-\left(\frac{a_1}{\delta}\right)^h}{1-\frac{a_1}{\delta}}\lambda^E_t + a_2 a_1^h \lambda^E_t + a_3 a_1^h \lambda^F_t$$

Then, after simplification:

$$\begin{aligned} -\frac{\kappa a_1}{\sigma\gamma} \sum_{h=0}^{\infty} \beta^h \mathbb{E}_t(\lambda^A_{t+h} - \lambda^A_{t+h-1}) = \\ \left[\frac{\kappa(1-\delta)a_1 a_2}{\sigma\gamma(1-\beta a_1)(1-\beta\delta)} - \frac{\kappa a_1(a_2+a_3)}{\sigma\gamma(1-\beta a_1)} \right] \chi\phi_t = (-\chi_4 - \chi_5)\chi\phi_t \end{aligned}$$

Plugging everything back into equation 7 and matching coefficients one last time gives the χ coefficient (i.e., the effect of a unit fiscal shock on the multipliers on impact) as:

$$\chi = [\chi_1 + \chi_2 + \chi_3 + \chi_4 + \chi_5]^{-1} \tag{58}$$

This pins down the paths for λ^E_t and λ^F_t, which then determine the optimal path for the remaining endogenous variables.

Appendix B: Log-linear approximation of the equilibrium conditions in the model with capital

The system of equilibrium conditions consists of the Euler equations for bonds and capital (17 and 18), the equations for the wage and rental rates (26 and 27, which imply market clearing in factor markets), the definition of marginal cost (eq. 23), the reset price equation (28), the definitions of the price level P_t and aggregate price distortions Δ_t, the market clearing condition for goods markets ($Y_t = C_t + I_t$), and the law of motion for capital ($K_t = I_t + (1-\tau)K_{t-1}$).

This system is log-linearised around a steady state with zero inflation, and for simplicity the productivity parameter is set to $A_t = 1$ for all t. In steady state, the real interest rate and the saving rate are:

$$1 + \bar{r}^k - \tau = \beta^{-1}$$

$$\bar{s} = \frac{\bar{I}}{\bar{Y}} = \tau \frac{\bar{K}}{\bar{Y}} = \frac{\alpha\beta\tau}{1-\beta+\beta\tau}$$

The log-linear approximation to the Euler equations (eq. 31) is standard, with r_t denoting the real ex-ante interest rate (purely an auxiliary variable in our model) and the usual Fisher equation being split in two (eqs. 35 and 36).

To log-linearize the additive equations for Y_t and K_t, use the log-linear approximation $a + b \approx \frac{\bar{a}}{\bar{a}+\bar{b}}\hat{a} + \frac{\bar{b}}{\bar{a}+\bar{b}}\hat{b}$, where bars represent steady state values and hats denote log deviations from steady state. Then, using the steady state values shown above yields 32 and 33.

For the factor markets block, first substitute wages out of 27. The resulting expression for the rental rate is:

$$r_t^k = \frac{\alpha}{1-\alpha} \frac{\Delta_t^{\left(1+\frac{1}{\psi}\right)\frac{1}{1-\alpha}} Y_t^{\frac{1}{\sigma}+\left(1+\frac{1}{\psi}\right)\frac{1}{1-\alpha}}}{K_{t-1}^{\left(1+\frac{\alpha}{\psi}\right)\frac{1}{1-\alpha}}} \left(\frac{C_t}{Y_t}\right)^{\frac{1}{\sigma}}$$

Then define $\hat{r}_t^k := \ln(1 + r_t^k - \tau) - \ln(1 + \bar{r}^k - \tau) \approx \frac{\bar{r}_t^k}{1+\bar{r}_t^k-\tau} \ln \frac{r_t^k}{\bar{r}_t^k}$ – that is to say, $\hat{r}_t^k$ is the log deviation of the gross return on capital, net of depreciation –, notice that $\frac{\bar{r}_t^k}{1+\bar{r}_t^k-\tau} = 1 - \beta(1-\tau)$, and take logs of the equation above to

obtain the log-linear approximation for the rental rate (equation 34). Note that $\ln \Delta_t \approx 0$ to a first order approximation.

To obtain the marginal cost equation, notice that equilibrium in the rental market for capital (equation 27) implies:

$$w_t = \frac{1-\alpha}{\alpha} r_t^k \left(\frac{\Delta_t Y_t}{K_{t-1}} \right)^{\frac{-1}{1-\alpha}}$$

Substituting this into the expression for marginal cost 23 gives us:

$$x_t = \alpha^{-1} r_t^k \frac{K_{t-1}}{\Delta_t Y_t}$$

Dividing by steady state values and taking logs then gives us equation 30.

Finally, the proof that the reset price equation 28 can be approximated by the New Keynesian Phillips curve 29 follows exactly the derivation of equation (16) in sections 3.2 and 3.3 of Galí (2015), with the only difference being that our model exhibits constant returns to scale, so the parameter α in Galí's notation should be set equal to 0.